Exploratory Data Analysis of a Three-Class Morphometric Dataset

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Abstract

We explore a 342-sample dataset of three classes (151 / 123 / 68 samples), with four continuous, unlabelled features and a sex attribute, before any modelling. Summary statistics, interquartile ranges, and univariate, bivariate and multivariate plots show three things. Class B is separable from the other two on a single feature (Feature 2 or Feature 3). Classes A and C differ almost only in Feature 1. And Features 3 and 4 are strongly correlated ($r = 0.87$), so they carry largely redundant information.

*The most instructive finding is a **Simpson's paradox**: Features 1 and 2 are* negatively *correlated across the whole dataset ($r = -0.24$), but* positively *correlated within every class ($r = 0.39$, 0.64 and 0.65). Pooled correlations would mislead any analysis that ignores class. The class sizes and feature ranges closely match a well-known bird-morphometrics dataset, which we note as a likely, but unconfirmed, identification.*

1. Introduction

Exploratory data analysis (EDA) [4] answers basic questions before any model is fitted: how balanced the classes are, how each feature is distributed, which features separate the classes, and which are redundant. Answering them early prevents the most common modelling errors, such as trusting a correlation that reverses within subgroups. We work through these questions systematically.

2. Related Work

Tukey [4] established EDA as a discipline built around robust summaries such as the median, quartiles and box plots. Kernel density estimates [2] underlie the violin and contour plots used here. Simpson's paradox [3], where an association reverses after conditioning on a grouping variable, is the classic warning against pooling heterogeneous groups.

3. Method

Using pandas and seaborn [5], we compute: (i) class frequencies and a class-by-sex table; (ii) summary statistics and interquartile ranges (IQR); (iii) univariate plots: a histogram of Feature 1 for class A, a box plot of Feature 2 and a violin plot of Feature 3 by class; (iv) bivariate plots: a Feature 1 vs. Feature 3 scatter, kernel-density contours of Feature 1 vs. Feature 4, and a hexagonal-bin plot of Feature 2 vs. Feature 4 for class A; (v) the Pearson correlation matrix, overall and within each class; and (vi) a pair plot of all four features coloured by class.

Table 1. Summary statistics of the four features ($n = 342$).

	Feature 1	Feature 2	Feature 3	Feature 4
Mean	43.92	17.15	200.92	4201.8
Std	5.46	1.97	14.06	802.0
Min	32.10	13.10	172.0	2700
Q1	39.23	15.60	190.0	3550
Median	44.45	17.30	197.0	4050
Q3	48.50	18.70	213.0	4750
Max	59.60	21.50	231.0	6300
IQR	**9.28**	**3.10**	**23.0**	**1200**

Table 2. Per-class mean (standard deviation).

	A ($n = 151$)	B ($n = 123$)	C ($n = 68$)
Feature 1	38.8 (2.7)	47.5 (3.1)	48.8 (3.3)
Feature 2	18.3 (1.2)	**15.0** (1.0)	18.4 (1.1)
Feature 3	190.0 (6.5)	**217.2** (6.5)	195.8 (7.1)
Feature 4	3701 (459)	**5076** (504)	3733 (384)

4. Results and Discussion

4.1. Data overview

Classes A, B and C have 151, 123 and 68 samples, so C is under-represented (≈20%). Sex is balanced within each class. Nine sex values are missing (8 blank, 1 recorded as "."); the four features are complete.

The features differ in scale by two orders of magnitude (Table 1), so distance-based models need standardisation. In every feature the mean is close to the median, which indicates fairly symmetric distributions. Feature 3 and Feature 4 are mildly right-skewed (mean above median).

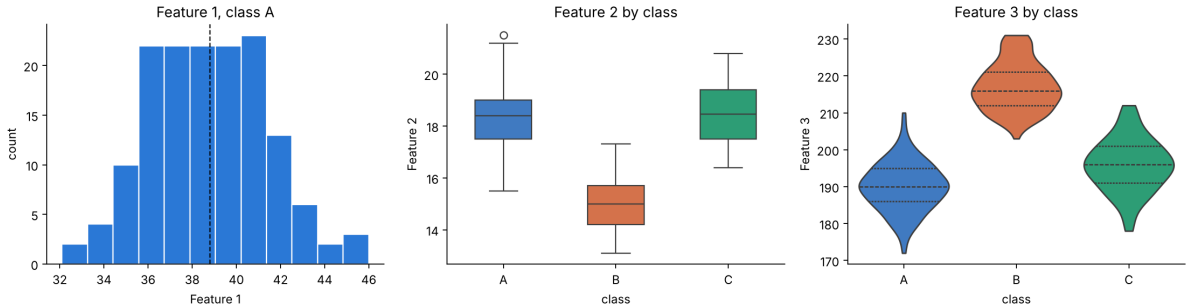


Figure 1. Left: Feature 1 for class A (dashed: mean). Centre: Feature 2 by class. Right: Feature 3 by class, with quartiles marked inside each violin.

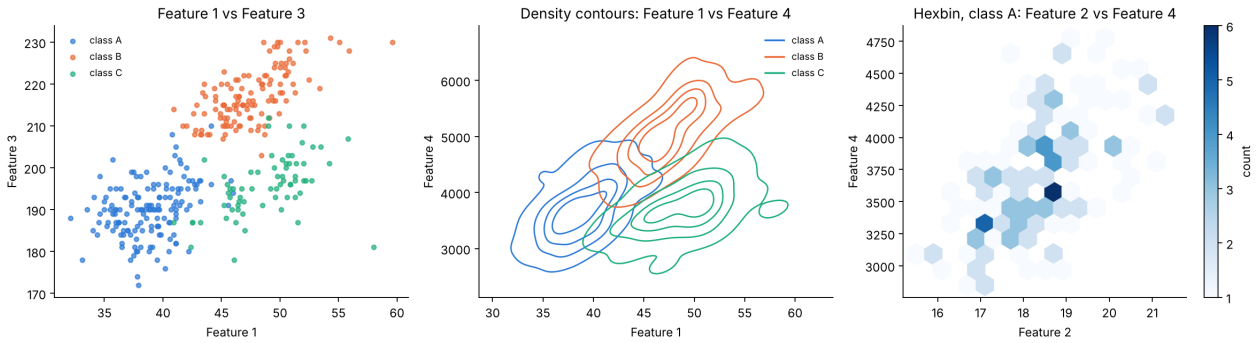


Figure 2. Left: Feature 1 vs. Feature 3. Centre: density contours of Feature 1 vs. Feature 4 per class. Right: hexagonal-bin counts of Feature 2 vs. Feature 4 for class A.

4.2. Univariate views

Feature 1 for class A is unimodal and roughly symmetric around 38.8 (Fig. 1). The box plot shows class B with a distinctly lower Feature 2: B's maximum (17.3) is below the means of A and C (18.3 and 18.4). The violin plot shows B's Feature 3 lying almost entirely above A and C (B minimum 203 vs. A/C maximum 212). Either feature alone nearly isolates class B.

4.3. Bivariate and multivariate views

In Feature 1 vs. Feature 3 (Fig. 2) the three classes form three nearly disjoint groups: A at low values of both, B at high Feature 3, and C at high Feature 1 but low Feature 3. Together these two features separate all three classes, which no single feature does. The Feature 1 / Feature 4 contours show the same pattern, with B's density displaced upward. For class A, the hexbin plot shows a positive Feature 2–Feature 4 relationship ($r = 0.58$), densest near the class medians (18.4, 3,700).

Redundancy. Features 3 and 4 are strongly correlated ($r = 0.87$, Fig. 3). Part of this comes from class differences, since within classes r is 0.47–0.70, but the two features still largely duplicate each other. Feature 2 correlates negatively with every other feature.

Simpson's paradox. The pooled correlation of Features 1 and 2 is -0.24, but within classes it is $+0.39$ (A), $+0.64$ (B) and $+0.65$ (C). The negative pooled value is an artefact of class B combining high Feature 1 with low Feature 2: the *between*-class difference outweighs the *within*-class relationship. The pair plot (Fig. 4) shows this directly, with each class's cloud tilting upward while the overall cloud tilts down.



Figure 3. Pearson correlations computed over all 342 samples.

A likely identity for the data. The class sizes (151/123/68), the feature ranges (e.g. 32–60, 13–22, 172–231, 2,700–6,300) and the per-class patterns match the Palmer Penguins dataset [1] (bill length and depth in mm, flipper length in

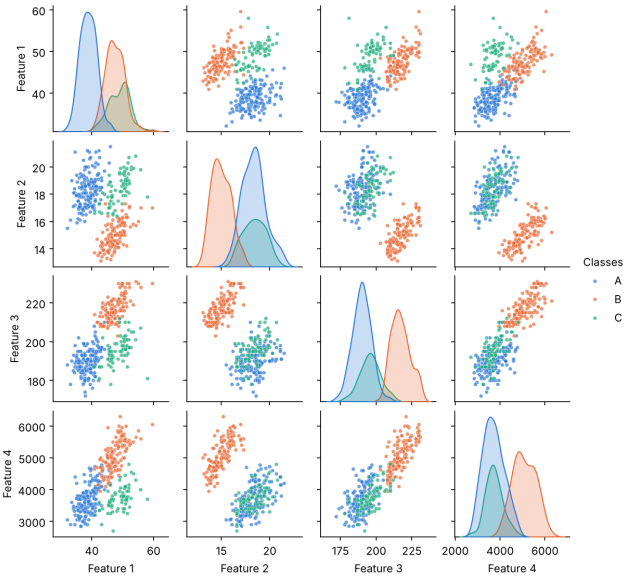


Figure 4. Pair plot of the four features by class (diagonal: per-class density).

mm, and body mass in g for Adelie, Gentoo and Chinstrap penguins) after removal of incomplete rows. This is an inference from the statistics; the file itself gives no feature names. Within every class, males are heavier than females by 400–800 in Feature 4, consistent with this reading.

5. Conclusion & Future Work

Two features (Feature 1 and Feature 3, or Feature 1 and Feature 2) are enough to separate all three classes, Feature 4 largely duplicates Feature 3, and pooled correlations must be interpreted per class because of a clear Simpson's paradox. **Next steps.** (i) Quantify separability with a simple classifier (e.g. logistic regression on Features 1 and 3); (ii) project the data with PCA or LDA to confirm the class geometry; (iii) handle the missing and invalid sex entries explicitly; and (iv) include sex as a covariate, since it shifts Feature 4 within every class.

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References

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